

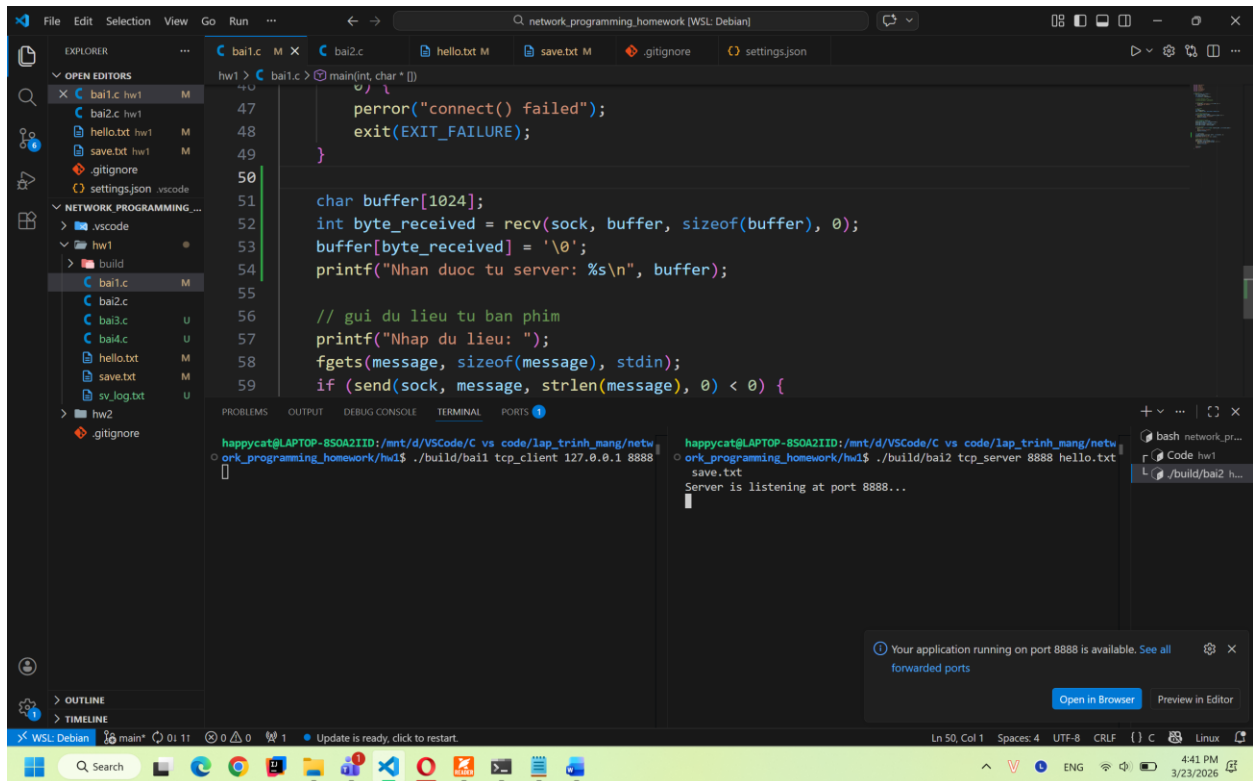
BÁO CÁO BÀI TẬP VỀ NHÀ SỐ 1 (NGÀY 17/03/2026)

HỌ VÀ TÊN: PHAN MINH VƯỢNG

MSSV: 20225241

1. Bài 1, 2 (tcp_client, tcp_server)

- Chạy server ở cổng 8888, file câu chào: hello.txt, file lưu: save.txt



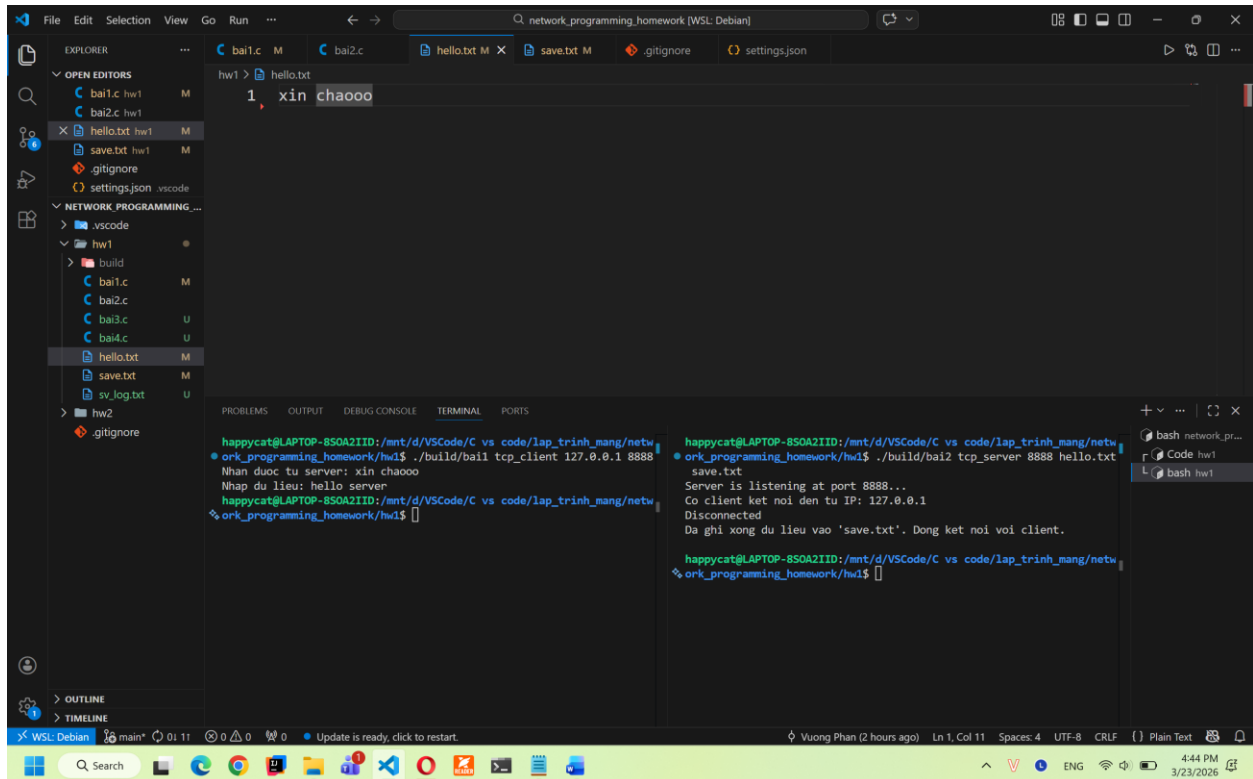
The screenshot shows a Visual Studio Code editor window with a project named "network_programming_homework [WSL: Debian]". The Explorer sidebar on the left shows the file structure, including folders "hw1" and "hw2", and files like "bai1.c", "bai2.c", "hello.txt", "save.txt", ".gitignore", and "settings.json".

The main editor displays the source code for "bai1.c", which is a C program for a TCP server. The code includes headers for `stdio.h`, `stdlib.h`, `unistd.h`, `sys/types.h`, `sys/socket.h`, `netinet/in.h`, and `arpa/inet.h`. It defines a `main` function that takes an integer `argc` and a character pointer `argv`. The program sets up a socket, binds it to port 8888, and listens for incoming connections. It uses `recv` to receive data from the client and `printf` to display it. It also uses `fgets` to read input from the user and `send` to send a response back to the client.

The terminal at the bottom shows the execution of the program. The prompt is `happycat@LAPTOP-850A211D: /mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw1$`. The user runs `./build/bai1 tcp_client 127.0.0.1 8888`, and the output is `Server is listening at port 8888...`. Another terminal window shows the user running `./build/bai2 tcp_server 8888 hello.txt save.txt`, and the output is `Server is listening at port 8888...`.

A notification at the bottom right of the terminal area states: "Your application running on port 8888 is available. See all forwarded ports". There are buttons for "Open in Browser" and "Preview in Editor".

- File hello.txt

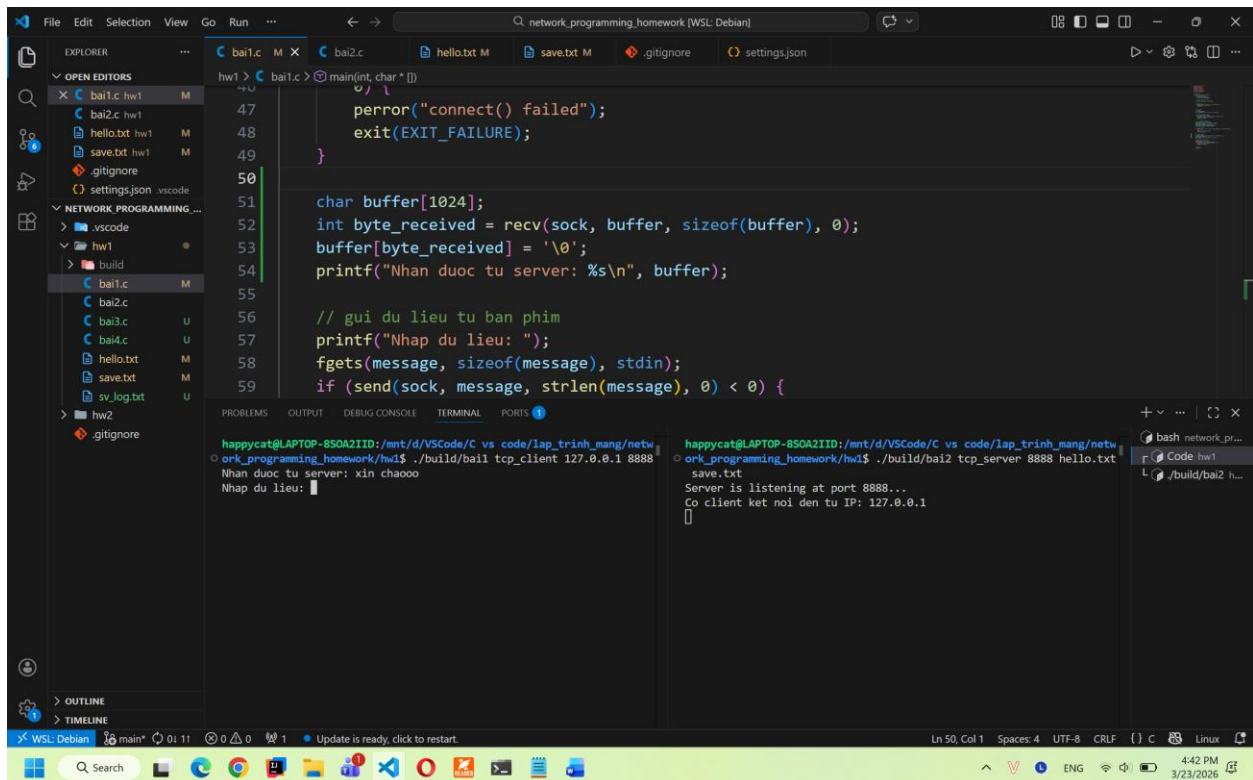


The screenshot shows the VS Code interface with the Explorer view on the left. The file explorer shows a project named 'network_programming_homework' with a subdirectory 'hw1'. Inside 'hw1', there are files 'bai1.c', 'bai2.c', 'hello.txt', 'save.txt', and '.gitignore'. The 'hello.txt' file is open in the editor, showing the text 'xin chao'. The Terminal view at the bottom shows a netcat listener on port 8888. It receives a connection from 127.0.0.1 and prints 'Nhan duoc tu server: xin chao'. The user then types 'xin chao' in the terminal, and the server responds with 'Server is listening at port 8888... Co client ket noi den tu IP: 127.0.0.1 Disconnected'.

```
happycat@LAPTOP-850A21ID: /mnt/d/VSCode/C vs code/lap_trinh_nang/netw
ork_programming_homework/hw1$ ./build/bai1 tcp_client 127.0.0.1 8888
Nhan duoc tu server: xin chao
Nhap du lieu: xin chao
happycat@LAPTOP-850A21ID: /mnt/d/VSCode/C vs code/lap_trinh_nang/netw
ork_programming_homework/hw1$
```

```
happycat@LAPTOP-850A21ID: /mnt/d/VSCode/C vs code/lap_trinh_nang/netw
ork_programming_homework/hw1$ ./build/bai2 tcp_server 8888 hello.txt
save.txt
Server is listening at port 8888...
Co client ket noi den tu IP: 127.0.0.1
Disconnected
Da ghi xong du lieu vao 'save.txt'. Dong ket noi voi client.
happycat@LAPTOP-850A21ID: /mnt/d/VSCode/C vs code/lap_trinh_nang/netw
ork_programming_homework/hw1$
```

- Client kết nối với server tại 127.0.0.1:8888 và nhận được xâu chào



The screenshot shows the VS Code interface with the Explorer view on the left. The file explorer shows the same project structure as the previous screenshot. The 'bai1.c' file is open in the editor, showing the source code for the client program. The code includes a main function that calls 'connect()' to connect to the server at 127.0.0.1:8888. It then receives data from the server and prints it. The code also includes a loop for sending data to the server. The Terminal view at the bottom shows the same netcat listener as the previous screenshot. It receives a connection from 127.0.0.1 and prints 'Nhan duoc tu server: xin chao'. The user then types 'xin chao' in the terminal, and the server responds with 'Server is listening at port 8888... Co client ket noi den tu IP: 127.0.0.1 Disconnected'.

```
happycat@LAPTOP-850A21ID: /mnt/d/VSCode/C vs code/lap_trinh_nang/netw
ork_programming_homework/hw1$ ./build/bai1 tcp_client 127.0.0.1 8888
Nhan duoc tu server: xin chao
Nhap du lieu: 
```

```
happycat@LAPTOP-850A21ID: /mnt/d/VSCode/C vs code/lap_trinh_nang/netw
ork_programming_homework/hw1$ ./build/bai2 tcp_server 8888 hello.txt
save.txt
Server is listening at port 8888...
Co client ket noi den tu IP: 127.0.0.1
Disconnected
```

- Client nhập dữ liệu và gửi đi, server nhận được và ghi vào save.txt

```
File Edit Selection View Go Run ... network_programming_homework [WSL: Debian]
EXPLORER
  bai1.c M
  bai2.c M
  hello.txt M
  save.txt M
  .gitignore
  settings.json
  .vscode
    hw1
      build
        bai1.c M
        bai2.c M
        bai3.c U
        bai4.c U
        hello.txt M
        save.txt M
        sv_log.txt U
      hw2
        .gitignore
  OUTLINE
  TIMELINE

main() {
    int sock = socket(AF_INET, SOCK_STREAM, 0);
    if (sock < 0) {
        perror("connect() failed");
        exit(EXIT_FAILURE);
    }

    char buffer[1024];
    int byte_received = recv(sock, buffer, sizeof(buffer), 0);
    buffer[byte_received] = '\0';
    printf("Nhan duoc tu server: %s\n", buffer);

    // gui du lieu tu ban phim
    printf("Nhap du lieu: ");
    fgets(message, sizeof(message), stdin);
    if (send(sock, message, strlen(message), 0) < 0) {
        perror("send failed");
        exit(EXIT_FAILURE);
    }
}

happyocat@LAPTOP-850A21ID:/mnt/d/VSCode/C vs code/lap_trinh_nang/netw
ork_programming_homework/hw1$ ./build/bai1 tcp_client 127.0.0.1 8888
Nhan duoc tu server: xin chaooo
Nhap du lieu: hello server
happyocat@LAPTOP-850A21ID:/mnt/d/VSCode/C vs code/lap_trinh_nang/netw
ork_programming_homework/hw1$

happyocat@LAPTOP-850A21ID:/mnt/d/VSCode/C vs code/lap_trinh_nang/netw
ork_programming_homework/hw1$ ./build/bai2 tcp_server 8888 hello.txt
save.txt
Server is listening at port 8888...
Co client ket noi den tu IP: 127.0.0.1
Disconnected
Da ghi xong du lieu vao 'save.txt'. Dong ket noi voi client.

happyocat@LAPTOP-850A21ID:/mnt/d/VSCode/C vs code/lap_trinh_nang/netw
ork_programming_homework/hw1$
```

- Kết quả trong file save.txt

```
File Edit Selection View Go Run ... network_programming_homework [WSL: Debian]
EXPLORER
  bai1.c M
  bai2.c M
  hello.txt M
  save.txt M
  .gitignore
  settings.json
  .vscode
    hw1
      build
        bai1.c M
        bai2.c M
        bai3.c U
        bai4.c U
        hello.txt M
        save.txt M
        sv_log.txt U
      hw2
        .gitignore
  OUTLINE
  TIMELINE

1 | hello server
2 |

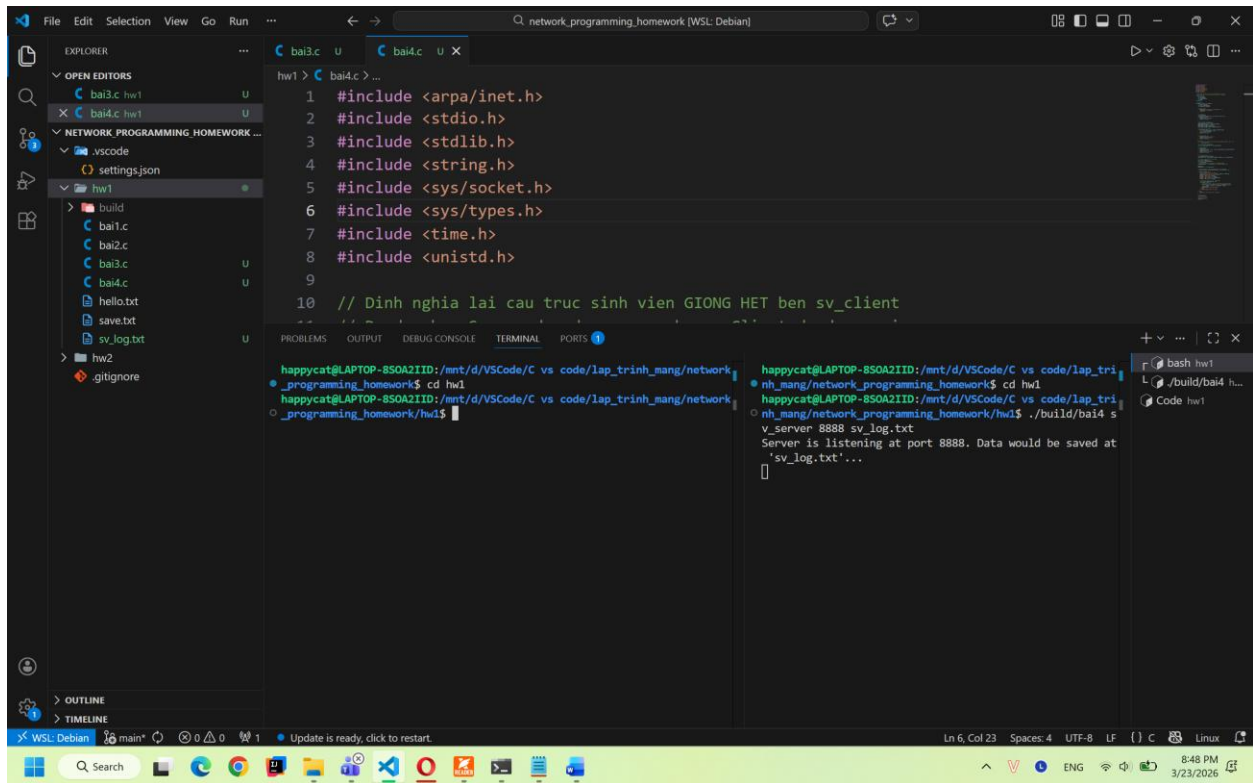
happyocat@LAPTOP-850A21ID:/mnt/d/VSCode/C vs code/lap_trinh_nang/netw
ork_programming_homework/hw1$ ./build/bai1 tcp_client 127.0.0.1 8888
Nhan duoc tu server: xin chaooo
Nhap du lieu: hello server
happyocat@LAPTOP-850A21ID:/mnt/d/VSCode/C vs code/lap_trinh_nang/netw
ork_programming_homework/hw1$

happyocat@LAPTOP-850A21ID:/mnt/d/VSCode/C vs code/lap_trinh_nang/netw
ork_programming_homework/hw1$ ./build/bai2 tcp_server 8888 hello.txt
save.txt
Server is listening at port 8888...
Co client ket noi den tu IP: 127.0.0.1
Disconnected
Da ghi xong du lieu vao 'save.txt'. Dong ket noi voi client.

happyocat@LAPTOP-850A21ID:/mnt/d/VSCode/C vs code/lap_trinh_nang/netw
ork_programming_homework/hw1$
```

2. Bài 3, 4 (sv_client, sv_server)

- Server mở ở cổng 8888, sẽ lưu vào file sv_log.txt



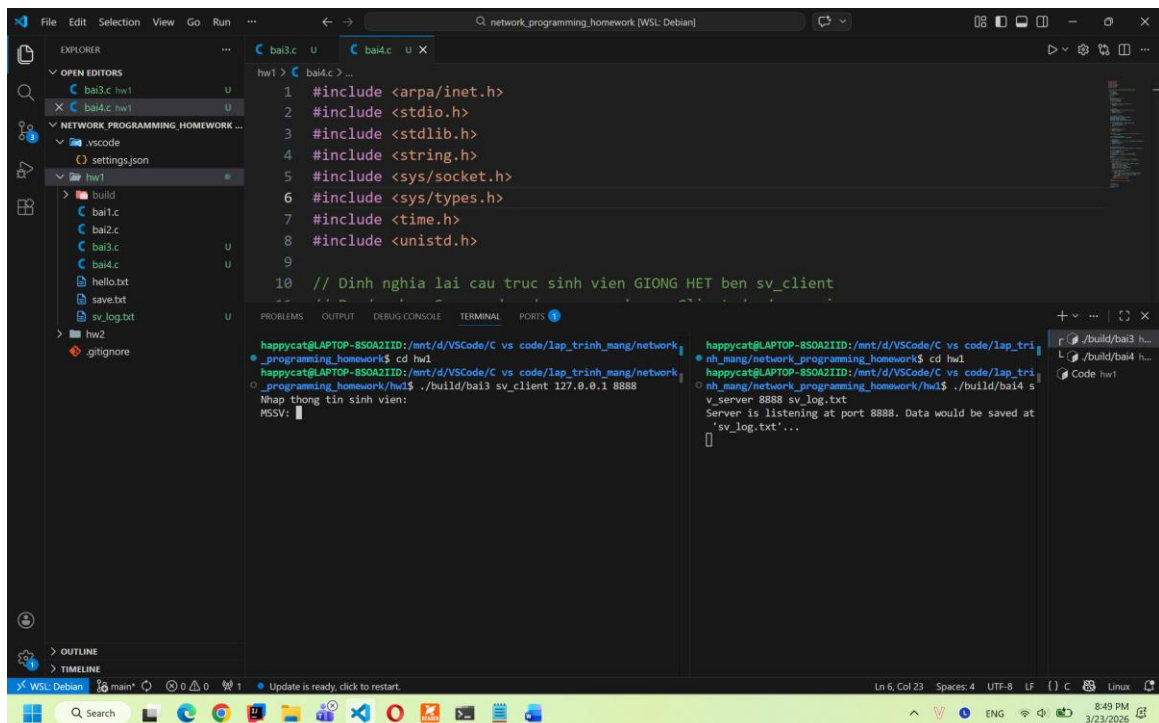
The screenshot shows the VS Code editor with a file explorer on the left displaying a project structure for 'network_programming_homework'. The main editor shows a C file named 'bai4.c' with the following code:

```
1 #include <arpa/inet.h>
2 #include <stdio.h>
3 #include <stdlib.h>
4 #include <string.h>
5 #include <sys/socket.h>
6 #include <sys/types.h>
7 #include <time.h>
8 #include <unistd.h>
9
10 // Định nghĩa lại cấu trúc sinh viên GIONG HET bên sv_client
```

The terminal at the bottom shows the execution of the server program:

```
happycat@LAPTOP-850A2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_
_programming_homework$ cd hw1
happycat@LAPTOP-850A2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_
_programming_homework/hw1$ ./build/bai4 s
v_server 8888 sv_log.txt
Server is listening at port 8888. Data would be saved at
'sv_log.txt'...
```

- Client kết nối vào 127.0.0.1:8888



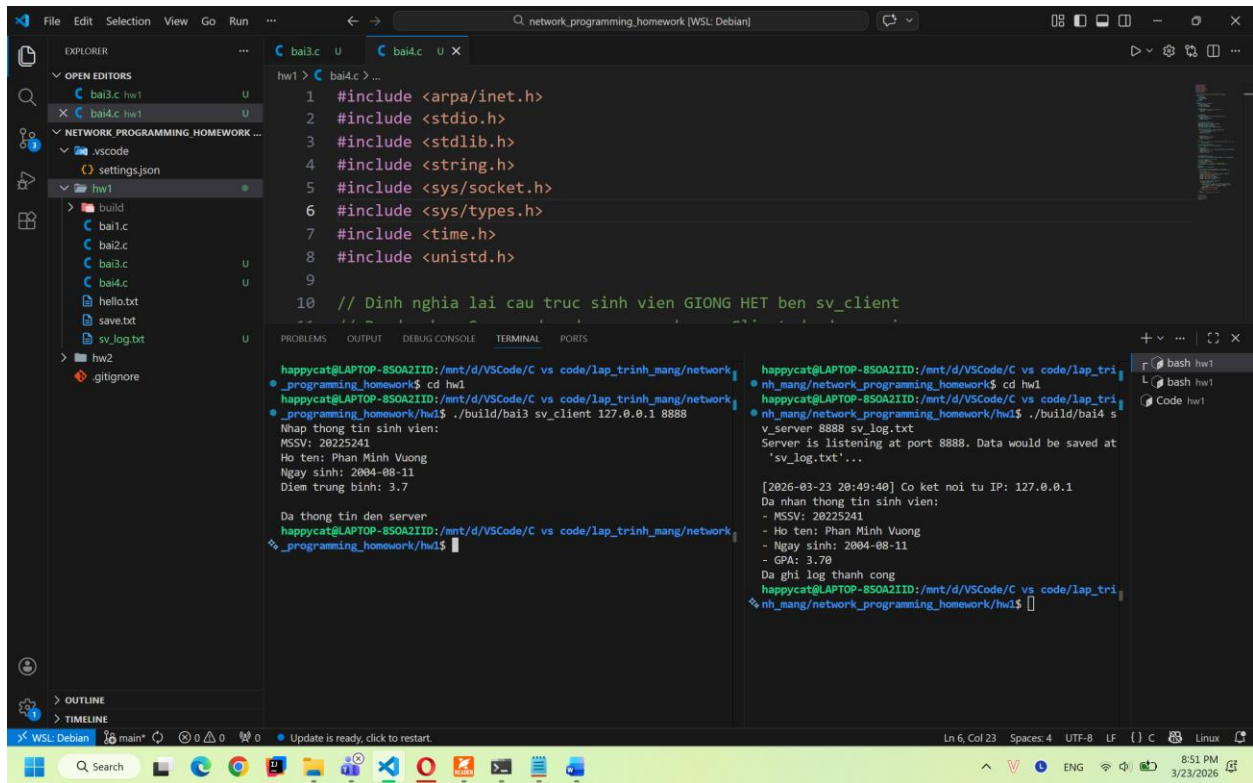
The screenshot shows the VS Code editor with the same project structure. The main editor shows a C file named 'bai3.c' with the following code:

```
1 #include <arpa/inet.h>
2 #include <stdio.h>
3 #include <stdlib.h>
4 #include <string.h>
5 #include <sys/socket.h>
6 #include <sys/types.h>
7 #include <time.h>
8 #include <unistd.h>
9
10 // Định nghĩa lại cấu trúc sinh viên GIONG HET bên sv_client
```

The terminal at the bottom shows the execution of the client program:

```
happycat@LAPTOP-850A2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_
_programming_homework$ cd hw1
happycat@LAPTOP-850A2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_
_programming_homework/hw1$ ./build/bai3 sv_client 127.0.0.1 8888
Nhập thông tin sinh viên:
MSSV: 
```

- Nhập thông tin sinh viên và gửi đến server, server nhận, in ra màn hình và ghi ra file sv_log.txt



The screenshot shows a Visual Studio Code editor window with a C program in `bai4.c`. The program includes headers for `arpa/inet.h`, `stdio.h`, `stdlib.h`, `string.h`, `sys/socket.h`, `sys/types.h`, `time.h`, and `unistd.h`. It defines a structure `GIOING_HET` and uses `sv_client`. The terminal shows the execution of the program, which prompts for student information and sends it to a server at `127.0.0.1:8888`. The server response is displayed in the terminal, and the data is also logged in `sv_log.txt`.

```
1 #include <arpa/inet.h>
2 #include <stdio.h>
3 #include <stdlib.h>
4 #include <string.h>
5 #include <sys/socket.h>
6 #include <sys/types.h>
7 #include <time.h>
8 #include <unistd.h>
9
10 // Định nghĩa lại cấu trúc sinh viên GIOING HET ben sv_client

happycat@LAPTOP-850A2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_
_programming_homework$ cd hw1
happycat@LAPTOP-850A2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_
_programming_homework/hw1$ ./build/bai3 sv_client 127.0.0.1 8888
Nhập thông tin sinh viên:
MSSV: 20225241
Họ tên: Phan Minh Vương
Ngày sinh: 2004-08-11
Điểm trung bình: 3.7

Đã thông tin đến server
happycat@LAPTOP-850A2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_
_programming_homework/hw1$

happycat@LAPTOP-850A2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_
nh_mang/network_programming_homework$ cd hw1
happycat@LAPTOP-850A2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_
nh_mang/network_programming_homework/hw1$ ./build/bai4 s
v_server 8888 sv_log.txt
Server is listening at port 8888. Data would be saved at
'sv_log.txt'...

[2026-03-23 20:49:40] Có kết nối từ IP: 127.0.0.1
Đã nhận thông tin sinh viên:
- MSSV: 20225241
- Họ tên: Phan Minh Vương
- Ngày sinh: 2004-08-11
- GPA: 3.70
Đã ghi log thành công
happycat@LAPTOP-850A2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_
nh_mang/network_programming_homework/hw1$
```

- Kết quả file sv_log.txt

